

13 Acute and Chronic Effects of Foam Rolling Effects on Range of Motion and Performance

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Stretching is not the only technique that can be used to enhance range of motion (ROM). Massage has been used since recorded history as a relaxation technique, cure for specific illnesses and performance enhancement. The first written records of the use of massage originated in Egypt and China. A Chinese text entitled, “The Yellow Emperor’s Classic Book of Internal Medicine.” was written around 2700 BCE. Egyptian tomb paintings around 2500 BCE illustrate massage as a medical therapy while written records of massage in India originate from 1500 BCE to 500 BCE. Western civilizations did not use massage as medical or health therapy until Per Ling from Sweden incorporated hand stroking into the Swedish Movement System in the early 1800s. Recently, devices such as foam rollers and roller massagers have become quite popular as self-massage therapy (see Figure 13.1) (1, 8, 18, 36, 37, 46, 52).

Fascial Restrictions and Myofascial Adhesions

Originally, foam rolling devices were branded as “self-myofascial release” devices that could aid in reducing (myo)fascial restrictions. Fascial restrictions are reported to occur in response to injury, disease, overuse, inactivity, and inflammation. Purportedly, if fascia loses its elasticity and becomes dehydrated, fascia or connective tissue can bind around the traumatized areas, causing a fibrous adhesion to form (49, 50). These myofascial adhesions may create “hypersensitive tender spots” (58), also known as trigger points. Fibrous adhesions can be painful, prevent normal muscle mechanics (i.e. neuromuscular hypertonicity, and decreased strength, endurance and motor coordination) and decrease soft-tissue extensibility negatively affecting joint ROM and muscle length (49, 50).

Type of foam rollers

Rolling involves small undulations back and forth over the affected muscle with a dense foam roller, typically starting at the proximal portion of the muscle, working down to the distal portion of the muscle or vice versa. Foam rollers are quite diverse in their composition with some made from polyvinyl chloride pipe surrounded by neoprene foam while others may be made from closed cell foam.



Figure 13.1 Foam rollers and roller massagers

Roller massagers are typically composed of dense foam wrapping around a solid plastic cylinder. There are many variations with some having a ridged design that is supposed to allow for both superficial and deep-tissue massage (8, 52). Besides the cylinder form, there are recent modifications such as foam ball rollers with various sizes, double foam ball rollers specially to treat the back extensors, or even small pointed pyramids to capture deep trigger points. Additionally, most of the devices are able to vibrate and there is a belief that this will increase the impact of the foam rolling exercise (48).

Acute foam rolling effects

A single session of foam rolling can increase the ROM of a joint immediately after the exercise (44, 60), and this increase can last for more than 60 minutes post-exercise (24). The latest techniques such as vibration foam rolling seem to increase the ROM of a joint to a greater extent compared to conventional foam rolling without vibration (45). Regarding the immediate impacts of foam rolling on various performance measures, a meta-analysis (59) suggests a trend towards enhanced sprint performance ($P = 0.06$; $+0.7$ percent; $ES = 0.28$), with minimal effects on jumping or strength performance. Additionally, reviews such as that by Cheatham et al. (10) corroborate findings indicating that a single bout of foam rolling does not significantly alter performance in the treated body regions. No changes in performance measures following a single bout of foam rolling can

actually be a positive response, as one of the major concerns about prolonged static stretching was the evidence for subsequent performance impairments (5, 6). According to meta-analyses there was no significant difference between the acute effects of stretching and foam rolling on ROM (28, 60) as well as on performance parameters (30). However, subgroup analyses revealed positive impacts of foam rolling on performance parameters when contrasted with static stretching, particularly when targeting specific muscles like the quadriceps or specific tasks such as strength exercises, especially when performed for durations exceeding 60 seconds or incorporating vibration. Conversely, when compared to dynamic stretching or utilized without vibration, similar effects were observed with no significant differences.

When combining foam rolling and stretching as a warm-up, a meta-analysis revealed a significant overall effect on ROM of the combined treatment when compared to no intervention (i.e., control condition). However, when a combined treatment (foam rolling + stretching) was compared to foam rolling or stretching alone a similar magnitude of change in ROM was shown (27). This indicates that a time efficient athlete can either stretch or perform foam rolling to acutely increase the ROM, as there seems to be no beneficial effect on ROM if those treatments are combined.

Chronic foam rolling effects

Regarding the long-term (i.e., training, chronic) effects of foam rolling, a recent meta-analysis from our research group (29) revealed that sustained foam rolling practice can lead to increased ROM over time ($ES = 0.82$), particularly when performed for more than four weeks and focused on the quadriceps and hamstrings. There was no significant improvement with the triceps surae however. One study showed that there is no additional effect when adding a vibration stimulus to the conventional foam rolling exercise (23). When it comes to performance parameters, another recent meta-analysis of our laboratory (26) suggests that foam rolling training does not result in significant changes ($ES = 0.294$; $p = 0.281$).

Rolling Mechanisms

So how does rolling increase ROM? The more obvious mechanisms exclusively for the acute increase in ROM are warm-up effects or thixotropy. The theory that simple warm-up effects are responsible for the acute changes in ROM is underlined by the literature. Our research group conducted an experiment where we compared a foam rolling exercise for 4x45 seconds on the hamstring muscles with a sham rolling 4x45 seconds by using a roller board to imitate the foam rolling movement. Interestingly, both groups (even the non-rolling sham group) increased the hip extension ROM similarly, which is very likely an indication for a global warm-up effect rather than changes in the local muscle properties (56). Similarly, a meta-analysis

reported no beneficial increases in ROM, owing to foam rolling (or stretching) when compared to other warm-up regimes such as cycling or running (57).

Another potential mechanism for an acute increase in ROM following a single bout of foam rolling are thixotropic effects as already discussed earlier. Thixotropic effects occur when viscous (thicker) fluids become less viscous or more fluid-like when agitated, sheared, or stressed. When the stress is removed, or desists then the fluid takes a certain period to return to the original viscous state. The rolling undulations place direct and sweeping pressure on the soft-tissue and generate friction. The increased friction-related temperature of the soft tissues and the stress from the rolling can decrease the viscosity of intracellular and extracellular fluid providing less resistance to movement. Remember the cold car and oil analogy.

The use of the term self-myofascial release suggests that rolling will help to release myofascial adhesions allowing freer less restricted movement. However, it is suggested that the amount of force needed to remove adhesions would exceed the physiological limitations of most people (49, 50). However, the research is not unanimous as it has been shown in one study that a single bout of foam rolling on the anterior thigh acutely released myofascial adhesions (while stretching does not) (34). Up to the present date, there is no evidence, if repeated foam rolling (i.e., training) treatments will release myofascial adhesions. However, as it was reported that a foam rolling training for more than four weeks can increase the ROM (29) one could speculate that at least one mechanism for this increase might be a chronic release in myofascial adhesions. However, are there mechanisms other than thixotropy that might explain changes in ROM following foam rolling?

Golgi tendon organs (GTO: type Ib afferents) responds to musculotendinous tension and strong stretch usually resulting in inhibition. A study from our laboratory (21) demonstrated that using a single bout of a short-duration (ten or 30 seconds) massage at the hamstrings musculotendinous junction increased ROM without an increase in passive muscle tension or electromyography (EMG) activity. Thirty seconds of musculotendinous massage provided greater overall ROM than 10 seconds. While one might assume that the tension applied to the tendon would activate the inhibitory responses of the GTO leading to greater muscle relaxation or decreased tonus, the GTO effects persist for only approximately 60 milliseconds after cessation of stress (54). Thus, the respective 5.8 percent–11.3 percent ROM increases immediately after the massage could not be due to GTO inhibition.

Muscle, fascia and skin are densely innervated by sensory neurons (49, 50). Within the epidermis and dermis (skin layers), there are mechanoreceptors such as the Merkel receptors, Meissner corpuscles, Ruffini cylinders, and Pacinian corpuscles which have different sized receptor fields, adapt slowly or rapidly and respond to different frequencies of skin stimulation. Merkel disks (small receptor field) and Ruffini cylinders (large receptor field) are slowly adapting and respond as long as the stimulus is present whereas the Meissner (small receptor field) and

Pacinian (large receptor field) corpuscles are rapidly adapting and thus respond to stimulation with a burst of firing activity at the beginning and end of stimulation. They respond to a variety of frequencies of stimulation ranging from 0.3–3 Hz (Merkel receptor), 3–40 Hz (Meissner corpuscle), 15–400 Hz (Ruffini cylinder) and 10–500 Hz (Pacinian corpuscle). So, all four mechanoreceptors would respond to slow rolling or massage with Ruffini cylinders and Pacinian corpuscles also responding to high frequency vibrations from machinery. Their major responsibilities are for proprioception and thus would not play a major role in neural inhibition, however Ruffini and Pacinian receptors may be able to contribute to the inhibition of sympathetic activity (contribute to muscle relaxation) (61). Ruffini receptors are more sensitive to tangential forces and lateral stretch (35), which would be prominent with rolling. Ruffini corpuscles stimulation may decrease sympathetic nervous system activity (55). Perhaps the increased relaxation, decreases in heart rate and blood pressure with massage can be partially attributed to contributions from manual stimulation of Ruffini and Pacinian receptors (58).

Another series of receptors that can affect sympathetic and parasympathetic activation are the interstitial type III and IV receptors. These receptors have both low and high threshold sensory capabilities. They are more numerous than type I and II receptors with either having a thin myelinated sheath or are unmyelinated and originate from free nerve endings. They are multi-modal receptors (many functions) responding to pain but also act as mechanoreceptors responding to tension and pressure. Their response to rapid and sustained pressure can contribute to decreases in heart rate, blood pressure, ventilation and lead to vasodilation (41). Thus, they can also contribute to a more relaxed muscle with less resistance to a full range of movement.

In terms of muscle stiffness, there are conflicting reports in the literature, although a current meta-analysis reported no acute changes in muscle stiffness following a single bout of foam rolling (12). Similarly in the long-term, various original studies reported no changes in muscle stiffness following several weeks of foam rolling training (23, 25).

Hence, there is growing evidence that mechanisms of rolling may be due to a number of factors including neural, psychological, and musculotendinous mechanisms. Magnusson has proposed that the acute or immediate increase in ROM following stretching is due to an increased stretch tolerance (38). In other words, the individual while stretching feels discomfort or pain and the act of holding that position allows him/her to diminish or accommodate the painful or uncomfortable sensation. If the same muscle length or joint ROM feels more tolerable then the person should be able to push themselves to an even greater ROM. There could be both psychological and neural components to this effect. Increased pain tolerance after foam rolling was confirmed following a single bout (22) as well as following a foam rolling training (23, 25). Consequently, pain modulation can be considered as a major component for the increase in ROM, owing to foam rolling.

Rolling can diminish different types of pain. Two studies from our lab punished subjects with ten sets of ten squat repetitions. The objective was to induce delayed onset muscle soreness (DOMS). In one study, participants were tested 24, and 48 hours after the squat protocol (46), whereas in the other they were also tested at 72 hours post-exercise (36). One of the groups would roll before testing and the other would not. In both studies, pain perception decreased when rolling was included post-exercise. DOMS-induced decreases in performance measures such as muscle activation, vertical jump, sprint time, and strength-endurance were attenuated (less impairments) with rolling. So, was pain decreased because of self-myofascial release or as a result of central (neural) mechanisms? Subsequent rolling and pain studies from our lab illustrate the importance of the neural component.

Whereas DOMS pain may endure for up to seven days owing to muscle damage and inflammatory responses (4), muscle tender points may persist for weeks and months. Based on the aforementioned studies, rolling can inhibit short duration pain (DOMS) but does it have an effect on chronic pain (i.e., muscle tender points or trigger points)? A study by Aboodarda et al. (1) had a massage therapist identify the most sensitive muscle tender point on participants' plantar flexors (calf). Then the calf was either massaged by the therapist, rolled or not touched (control). One more condition was included and that involved rolling of the contralateral calf. Both massaging and rolling the affected calf decreased the pain sensitivity of the muscle tender point. But even more interesting was that rolling the contralateral calf; *no need to even touch the painful calf*; reduced the pain sensitivity as well. In a similar study (9), we induced pain by electrically stimulating the tibial nerve of the plantar flexors with maximal and submaximal (70 percent of maximal current) intensity high frequency (50 Hz) electrical stimulation (tetanus). The experiment was not much fun for the participants as it involved maximal tolerable pain. Again, we either rolled the affected calf, did nothing (control) or rolled the contralateral calf. While each session of muscle stimulation increased pain sensitivity by about 10 percent, rolling the affected or contralateral calf inhibited the pain increase. Thus, if there is no need to touch the affected muscle and rolling another muscle decreases pain, then it is evident that some global neural response has occurred to decrease pain under chronic (muscle tender points), short term (DOMS) and acute (tetanic stimulation) pain conditions.

In the aforementioned Aboodarda et al. (1) pain depression occurred after light rolling massage. Nociceptors (pain receptors) are present in both muscle and skin (31, 32) and thus even light rolling can increase the sensitivity of superficial nociceptors. Related to this finding, another study from our laboratory (14) reported that the intensity of rolling did not differentially affect the ROM. That is, whether the roller massage was at 50 percent, 70 percent, or 90 percent of the maximum point of discomfort, the increase in ROM was similar.

There are a number of central nervous system possibilities including the gate control theory of pain (see Figure 9.7) (39, 42), diffuse noxious inhibitory control (see Figure 13.2) (40) and parasympathetic nervous system alterations.

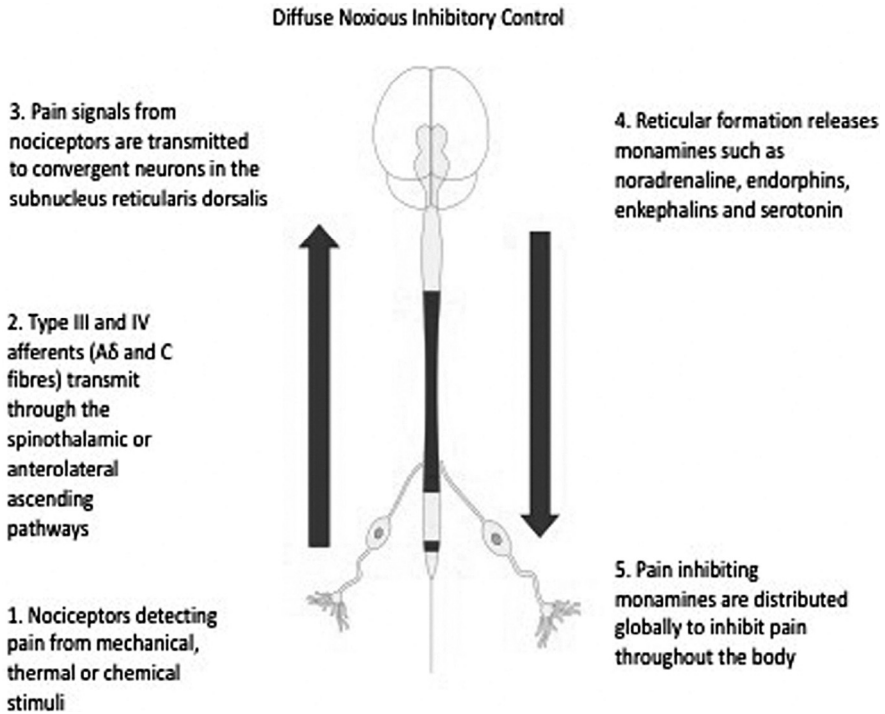


Figure 13.2 Diffuse noxious inhibitory control

The gate control theory of pain is commonly exemplified by somebody hurting a body part and immediately grabbing and rubbing that same body part to minimize the pain. This action would stimulate many receptors and afferents which would then be filtered at the brain. More technically the gate control theory involves activation of thick myelinated ergoreceptor (group III and IV afferents) nerve fibres (via activation of percutaneous [skin] and muscle mechanoreceptors, metaboreceptors and proprioceptors) that modify the signals from ascending nociceptors (pain receptors) via small diameter Aδ fibres to the periaqueductal grey nucleus (42). Analgesia (pain suppression) then results from descending signals to opioid receptors, which would inhibit pain with serotonergic and noradrenergic neurons (47).

Diffuse noxious inhibitory control (DNIC), also known as counter-irritation can be activated by nociceptive stimuli (i.e., mechanical pressure) from a non-local tissue. The typical practical application is that if you banged your head against the cupboard, you can stub (hurt) your toe and the head pain would subjectively decrease. With DNIC, activation of non-local receptors is transmitted to multi-receptive, wide dynamic range convergent neurons in the cortical subnucleus reticularis dorsalis where it inhibits pain transmission monoaminergically (i.e., using monoamine transmitters such as noradrenaline

and serotonin) reducing pain perception not only locally but also at distant sites (40, 47, 51).

Massage can also stimulate parasympathetic activation; with changes in serotonin, cortisol, endorphin, and oxytocin contributing to a decreased pain perception (58). Decreased parasympathetic reflexes could decrease pain sensitivity by reducing stress from myofascial tissue by relaxing the strain on the smooth muscles embedded in the soft tissue. Massaging or rolling the muscle can produce acute ischemic compression, which has been shown to result in reduced perceived pain in the upper trapezius muscles (33) and with neck and shoulder muscles (19).

Massage decreases the afferent excitability of the alpha motoneurons. A study from our laboratory found that both tapotement and petrissage massage (7) as well as roller massage techniques attenuated the H-reflex to M-wave ratio. Normalizing the H-reflex to the M-wave is an important technical procedure to ensure that any changes in the H-reflex are not due to alterations in the muscle's membrane action potential and are actually due to changes in the afferent excitability of the motoneuron. The suppression of the spinal reflex excitability with the H-reflex may be attributed to decreased alpha motoneuron excitability and/or increased presynaptic inhibition. A number of other studies have reported H-reflex inhibition with stretching (2, 11, 15, 16, 17) as well as massage (13, 53). Massage has suppressed the H-reflex by 40–90 percent in these studies.

Thus, while ROM increases with massage and rolling may be attributed to simple warm-up effects or thixotropic factors, a greater tolerance to the discomfort associated with musculotendinous lengthening (stretch tolerance), there could also be reflexive inhibition decreasing stretch-induced contractile activity resulting in a more relaxed muscle. Another potential but controversial mechanism for an increase in ROM might be the release of myofascial adhesions.

Other potential benefits or contradictors

When foam rolling the hamstrings, quadriceps, or calf muscles, it's important to assume a position akin to a plank. A prior study of our laboratory (62) indicated that the muscle engagement, as measured by EMG of the core muscles during a plank or reverse plank position is comparable to that during foam rolling of the quadriceps and hamstrings, respectively. Therefore, it's plausible to suggest that prolonged foam rolling of the front and back thigh muscles, as well as the calves, in a plank-like position could potentially enhance core strength and endurance over time.

Besides all the positive effects proposed in this chapter, potential contradictors have to be discussed. A Delphi study discussed possible reasons to avoid or proceed with caution when using foam rolling, none of which have been documented in clinical trials thus far. The panel of experts identified open wounds and bone fractures as reasons to avoid foam rolling, while conditions such as local tissue inflammation, deep

vein thrombosis, myositis ossificans, and osteomyelitis were deemed cautionary. The authors recommended that future research should focus on gathering evidence-based information on when to avoid or use caution with foam rolling for various conditions and in different professional settings (3).

Rolling Recommendations

What intensity and duration of rolling is optimal to achieve the greatest increases in ROM? Studies have demonstrated increased joint ROM with as little as 5–10 seconds of rolling with 10 seconds providing significantly greater increases than 5 seconds (52). Most studies use multiple sets of 30–60 seconds bouts of rolling and although not directly compared 60 seconds rolling seem to provide further ROM improvements (8, 18, 36, 37, 46). However, in one study, 60, 90 and 120 seconds of rolling were applied between four sets of knee extensions. While 120 seconds of rolling decreased the number of knee extension repetitions by 14 percent, the 90 and 60 seconds of rolling also decreased repetition numbers by 8–9 percent (43). This was one of the few studies that has reported impairments with rolling. As mentioned previously, whether rolling was conducted at 5/10, 7/10, or 9/10 on a pain scale, the increases in ROM were similar (14). Thus, there is no need to apply excruciating pain to achieve the desired results. Furthermore, following a typical warm-up (five minutes of aerobic activity, static and dynamic stretching and sport specific activities), if rolling is performed at ten-minute intervals after the warm-up, the augmented ROM is maintained for 30 minutes to a greater degree than without intermitted rolling (20). This finding would be very important for those athletes who do not start a game or match and must sit on a bench for 10–30 minutes before entering the game. Continuing to roll while waiting on the bench would help maintain some of the benefits of the warm-up. Illustrations of various foam rolling exercises can be found with Figures 13.3–13.27.



Figure 13.3 a,b: Gluteals

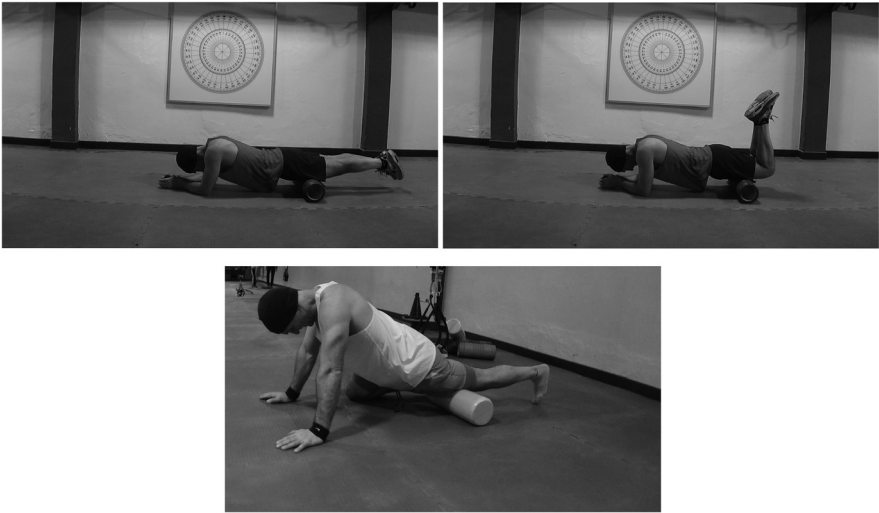


Figure 13.4 a,b,c: Quadriceps



Figure 13.5 Adductors (Groin)

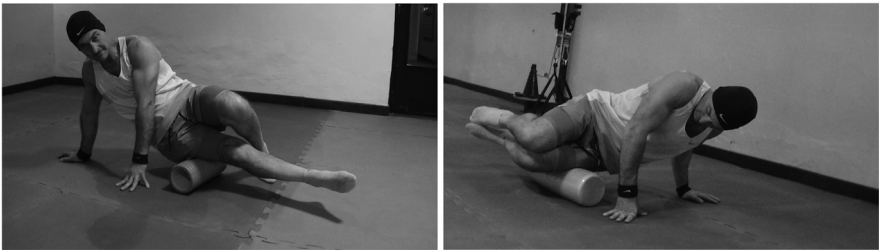


Figure 13.6 a,b: Tensor Fascia Latae



Figure 13.7 a,b: Hamstrings



Figure 13.8 Plantar Flexors

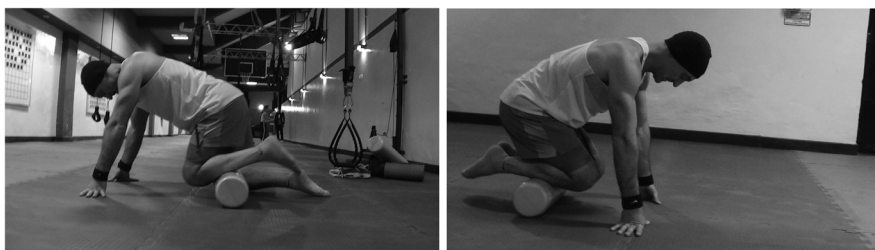


Figure 13.9 a,b: Tibialis Anterior



Figure 13.10 Inner calf

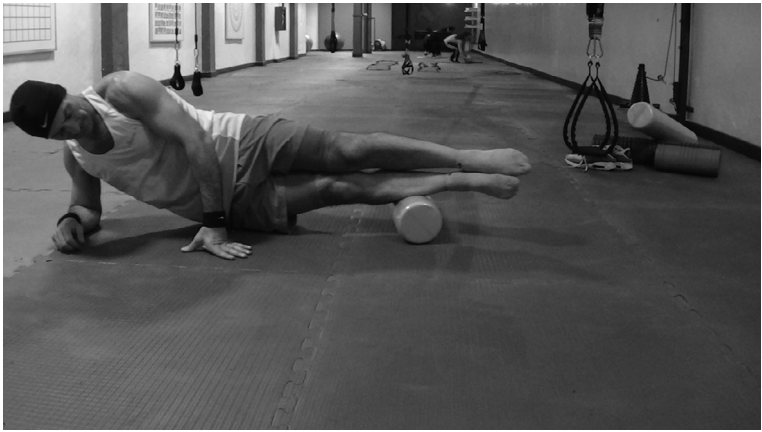


Figure 13.11 Outer calf



Figure 13.12 a,b: Sole of the foot



Figure 13.13 Pectorals

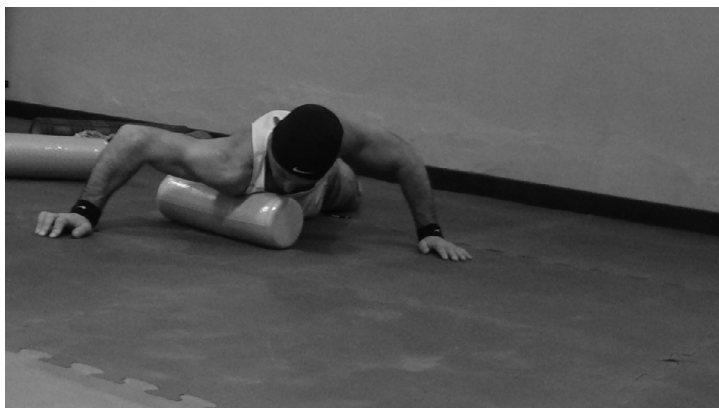


Figure 13.14 Pectorals and Anterior Deltoid



Figure 13.15 Lateral Shoulder (Deltoid) and Arm (Biceps and Triceps Brachii)



Figure 13.16 Posterior Shoulder (Deltoids) and Triceps Brachii



Figure 13.17 Posterior Shoulders (Deltoids and Trapezius)



Figure 13.18 Biceps Brachii



Figure 13.19 Triceps Brachii

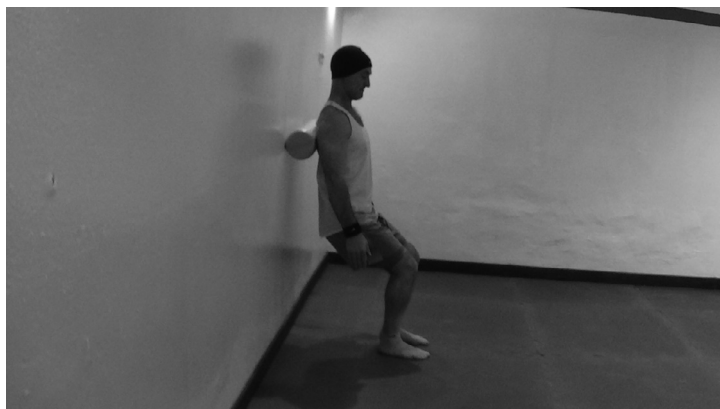


Figure 13.20 Back (Upright position)



Figure 13.21 Lower Back



Figure 13.22 a,b: Lateral Trunk



Figure 13.23 Neck



Figure 13.24 Gluteals and Plantar Flexors (Double Rolling)



Figure 13.25 Back and Arm (Double Rolling)



Figure 13.26 Roller Massage of the hamstrings.



Figure 13.27 Exercise Balls

Summary

Whereas massage has been used for millennia, foam rollers and roller massagers have enjoyed recent popularity. Rolling can increase flexibility for approximately 60 minutes without subsequent performance deficits. Mechanisms underlying these ROM changes may be ascribed to warm-up effects, decreased visco-elasticity (thixotropic effects), rolling-induced decreases in sympathetic activity and motoneuron excitability and increased stretch tolerance. It is recommended to perform multiple sets of 30–60 seconds of rolling, which can be performed below the maximum pain tolerance (50–90 percent of maximum pain tolerance).

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